Ecology, 96(2), 2015, pp. 318–324
© 2015 by the Ecological Society of America

Reports

A new probabilistic method for quantifying *n*-dimensional ecological niches and niche overlap

Heidi K. Swanson,[1,4] Martin Lysy,[2] Michael Power,[1] Ashley D. Stasko,[1] Jim D. Johnson,[3] and James D. Reist[3]

[1]*Department of Biology, University of Waterloo, 200 University Avenue West, Waterloo, Ontario N2L 3G1 Canada*
[2]*Department of Statistics and Actuarial Science, University of Waterloo, 200 University Avenue West, Waterloo, Ontario N2L 3G1 Canada*
[3]*Fisheries and Oceans Canada, 501 University Crescent, Winnipeg, Manitoba R3T 2N6 Canada*

Abstract. Considerable progress has been made in the development of statistical tools to quantify trophic relationships using stable isotope ratios, including tools that address size and overlap of isotopic niches. We build upon recent progress and propose a new probabilistic method for determining niche region and pairwise niche overlap that can be extended beyond two dimensions, provides directional estimates of niche overlap, accounts for species-specific distributions in niche space, and, unlike geometric methods, produces consistent and unique bivariate projections of multivariate data. We define the niche region (N_R) as a given 95% (or user-defined α) probability region in multivariate space. Overlap is calculated as the probability that an individual from species *A* is found in the N_R of species *B*. Uncertainty is accounted for in a Bayesian framework, and is the only aspect of the methodology that depends on sample size. Application is illustrated with three-dimensional stable isotope data, but practitioners could use any continuous indicator of ecological niche in any number of dimensions. We suggest that this represents an advance in our ability to quantify and compare ecological niches in a way that is more consistent with Hutchinson's concept of an "*n*-dimensional hypervolume."

Key words: arctic fish; Bayesian probabilistic analysis; Beaufort Sea, Yukon Territory, Canada; Coregonus *spp.; niche overlap; niche quantification; nicheROVER; stable isotopes.*

Introduction

Understanding differences in resource partitioning between organisms is central to ecology. Hutchinson's (1957) classic description of niches as "*n*-dimensional hypervolumes" has shaped much of the philosophy, direction, and methods in ecology during the last half-century. Often we are concerned with understanding overlap in niches, given that species may overlap along many niche axes but must differ in at least one to avoid competitive exclusion (Gause et al. 1934, Hutchinson 1957). Quantifying niches and niche overlap is of interest to many ecologists, and the use of stable isotope analysis in ecology has shaped much of the recent development in this area.

Stable isotope ratios (e.g., $\delta^{15}N$, $\delta^{13}C$) summarize trophic interactions for consumers, and can be used to calculate quantifiable "isotopic niches" (e.g., Bearhop et al. 2004, Newsome et al. 2007). Although proponents point out that the isotopic niche represents only a subset of the ecological niche, the concept of an isotopic niche has led to a proliferation of mathematical techniques for quantifying aspects of niche dimensionality (e.g., Layman et al. 2007, Jackson et al. 2011), intra- and interspecific diet variability (e.g., Bolnick et al. 2003, Araújo et al. 2007), source contributions from resource pools (e.g., Phillips and Gregg 2003, Moore and Semmens 2008), and niche variability across spatial and temporal gradients (e.g., Schmidt et al. 2009).

Stable isotope bi-plots (most commonly $\delta^{13}C$ and $\delta^{15}N$) have been qualitatively analyzed to infer food-web structure since the late 1970s. Since then, the use of stable isotopes in ecological studies has exploded, and researchers have employed a more diverse set of isotopic tracers, including $\delta^{18}O$, $\delta^{34}S$, and δD (deuterium). Bearhop et al. (2004) were the first to explicitly propose using variance in stable isotope ratios as a measure of trophic niche width that could be compared between populations or species using simple *F* tests. Layman et al. (2007) expanded on this idea, and proposed a set of metrics to quantify several aspects of niche dimensionality. Total area (TA) of a convex hull drawn around the outermost values in an isotope bi-plot was suggested as a measure of trophic niche space occupied by the population or community. Concerns regarding the sensitivity of convex hulls to sample size and outliers prompted Jackson et al. (2011) to suggest using more robust standard ellipse areas (SEA) as a boundary for isotopic niche space. This concept leads to a geometric

Manuscript received 3 February 2014; revised 9 October 2014; accepted 14 October 2014. Corresponding Editor: B. D. Inouye.

[4] E-mail: hswanson@uwaterloo.ca

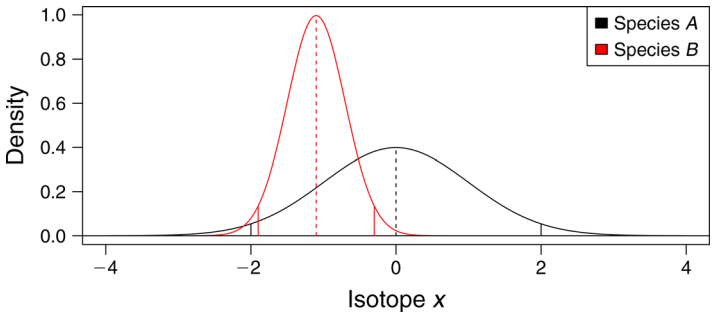


FIG. 1. Comparison between geometric and probabilistic isotope overlap metrics. Isotope x is distributed in species A and B (x-axis); dashed lines show means. The 95% niche region (solid lines) of B is completely contained within that of A, such that the geometric overlap of B onto A is 100%. However, the probability that a randomly chosen member from A falls into the isotopic niche of B is only 35%.

calculation of the area of overlap between two standard ellipses (see Parnell and Jackson 2013), which has been adopted as a proxy for isotopic niche overlap (e.g., Jackson et al. 2012, Guzzo et al. 2013).

The bivariate niche models developed to suit stable isotope data (e.g., Bearhop et al. 2004, Layman et al. 2007, Newsome et al. 2007, Jackson et al. 2011) have a more efficient and intuitive view of trophic niche space than traditional multivariate analyses (e.g., Green 1971, Pianka 1974, Petraitis 1979). The bivariate isotopic niche framework, however, continues to suffer from two major drawbacks. First, geometric concepts of niche do not generalize well to higher dimensions. The notion of niche "area" (or "volume") ceases to be intuitive for $n > 3$ dimensions, and it is unclear how an n-dimensional hypervolume is to be projected into two dimensions for visual display. Second, a purely geometric notion of trophic niche overlap can be somewhat problematic. For instance, consider the area of intersection between a pair of two-dimensional niches. While this notion lacks directionality (overlap between species A and B as opposed to overlap of A on B), this can easily be resolved by defining overlap as the fraction of the intersection area between niche A and niche B over the total niche area of B (and vice versa). However, such a computation assumes an even distribution of organisms within the defined niche spaces. Concerns arising from an assumption of constant resource availability and selectivity have a long history in trophic niche overlap metrics that predates the use of stable isotope ratios to define trophic niche space (e.g., Colwell and Futuyma 1971, Petraitis 1979). In fact, the magnitude of niche overlap defined geometrically might have a counterintuitive interpretation, as illustrated by the following example.

Fig. 1 displays the distribution of a single isotope x in two species A and B. Using geometric methods, the isotopic niche of species B is completely overlapped by species A. However, the probability of finding an individual of species A in species B's niche is only 35%. This still may be ecologically relevant (that is up to the investigator to decide), but it is important to recognize that not all individuals of species A will share isotopic niche space with species B.

In this study, we describe and demonstrate a new and simple analytical approach for quantifying niche size and overlap. The former is defined as a probability region (95% in our example) in n-dimensional space and reduces to elliptical regions for $n = 2$. The latter is defined as the probability of one species falling into the niche of another, as in Fig. 1. We apply the method to a case with four fish species and $n = 3$ niche axes. Our proposed approach is not restricted to stable isotope data (or to $n = 3$ axes), but we chose an example using three isotope tracers because of the considerable momentum in this discipline. We suggest that the use of probabilities for examining niche overlap simplifies analyses both intuitively and computationally, especially when the niche is defined using $n > 2$ dimensions. By adopting a Bayesian inference framework, all of our estimates naturally incorporate a measure of uncertainty. In proposing this method, we hope to build upon the ideas and analytical methods previously developed for stable isotope ecology, and make them relevant and accessible to a wider set of ecological questions.

STATISTICAL METHODS

Probabilistic niche regions

Let $X = (X_1, \ldots, X_n)$ denote the n-dimensional stable isotope measurements for a randomly chosen individual from a given species. In our probabilistic setting, we define the species' trophic niche region N_R as a specific region of n-dimensional space in which X has probability α of being found. We use $\alpha = 95\%$, but the accompanying software (nicheROVER) allows α to be set at any value ($\alpha = 95\%$ is default).

As there are infinitely many regions of space for which we have 95% probability of finding X, we let N_R be (in some sense) the smallest one. This turns out to be the 95% region consisting of the most likely values of X (e.g., Box and Tiao 1973). That is, let $f(x)$ denote the joint probability density function (joint pdf) of X. Then a value x_1 is more likely than x_2 if $f(x_1) > f(x_2)$. Thus, we define N_R as

$$N_R = \{x : f(x) > C\} \quad (1)$$

where the constant C is chosen such that $\Pr(X \in N_R) = 95\%$ (details and illustration in Appendix A).

In the n-dimensional setting, it is of particular interest to determine N_R under the assumption that $X \sim \mathcal{N}(\mu, \Sigma)$ is multivariate normal, with mean μ and variance Σ.

19399170, 2015, 2, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/14-0235.1 by Universidade Do Algarve, Wiley Online Library on [27/10/2022]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

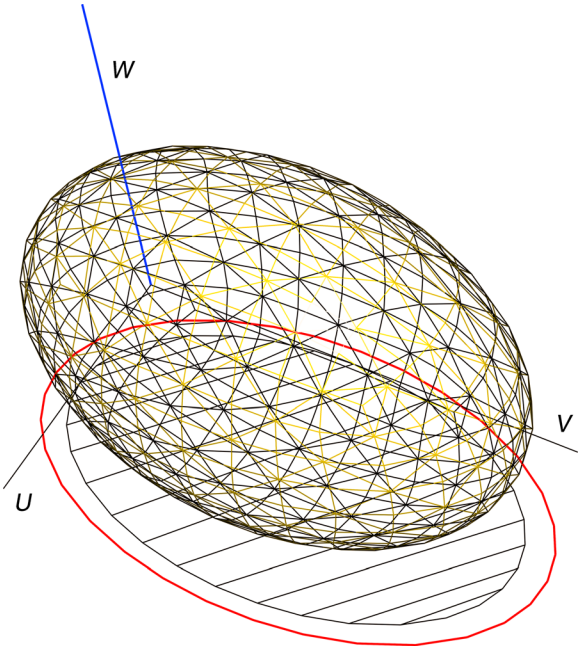


FIG. 2. A three-dimensional 95% niche region for normally distributed isotopes. The two ellipses are projections of the niche region (N_R) onto isotope axes U and V. The black, hatched ellipse is the geometric projection of N_R onto this plane, which is the shadow obtained by shining a light onto the three-dimensional ellipsoid along the W-axis (in blue). This is the largest of the two-dimensional ellipse "slices" taken parallel to the UV-plane. The red ellipse is the 95% niche region constructed directly from isotopes U and V. Thus, the geometric projection of the ellipsoid covers less than 95% of randomly selected individuals in two-dimensional isotope space.

When this is the case, the points in N_R satisfy a quadratic inequality,

$$N_R = \{x = (x_1, \ldots, x_n) : (x - \mu)'\Sigma^{-1}(x - \mu) < C\}$$

$$= \left\{x = (x_1, \ldots, x_n) : \sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}x_ix_j + \sum_{i=1}^{n} b_ix_i + d < C\right\} \quad (2)$$

with polynomial coefficients a_{ij}, b_i, and d being readily computable from the values of μ and Σ (Appendix A).

For a single isotope ($n = 1$), the niche region for normally distributed data is simply the interval $\mu \pm 1.96 \times \sigma$. For bivariate isotopes ($n = 2$), N_R reduces to the Bayesian elliptical niche regions proposed by Jackson et al. (2011). For higher dimensions ($n > 2$), N_R can be difficult to visualize. However, it is easy to "project" the niche region onto any two axes for visual display. For instance, with $n = 4$ isotopes T, U, V, W, the projection of N_R onto isotopes U and V is simply the two-dimensional niche region calculated with U and V. Note that this is a probabilistic projection which does not coincide with the geometric projection of N_R onto the UV-plane. In fact, the geometric approach produces different two-dimensional projections depending on the total number of isotopes n, whereas the probabilistic projection is unique. An illustration of the two different projections for $n = 3$ is displayed in Fig. 2.

Niche overlap

Now suppose that X_A and X_B correspond to randomly selected individuals from species A and species B, and let $N_R(A)$ and $N_R(B)$ denote their respective niche regions. We define the niche overlap of species A onto species B as the probability that an individual from species A is found in the niche of B

$$O(^A_B) = \Pr(X_A \in N_R(B)). \quad (3)$$

Even for normally distributed isotopes within each species, calculation of this overlap metric involves an n-dimensional integral which is difficult to evaluate directly. However, it can easily be approximated by the following Monte Carlo technique (full details in Appendix B). First, we simulate M independent and identically distributed (iid) draws $X_A^{(1)}, \ldots, X_A^{(M)}$ from the distribution $\mathcal{N}(\mu_A, \Sigma_A)$ of species A. For each $X_A^{(m)}$, we define an indicator variable I_m such that $I_m = 1$ if $X_A^{(m)}$ is inside $N_R(B)$, and $I_m = 0$ otherwise. This amounts to checking whether or not $X_A^{(m)}$ satisfies the quadratic inequality (2), with coefficients determined by μ_B and Σ_B. Finally, the overlap metric is approximated by $\hat{O}(^A_B) = \frac{1}{M}\sum_{m=1}^{M} I_m$, providing an unbiased estimate of trophic niche overlap with variance decreasing to 0 as $M \to \infty$.

Parameter estimation

While it is often justifiable to assume that stable isotope distributions are multivariate normal (e.g., Jackson et al. 2011), the mean and variance parameters μ and Σ are unknown and must be estimated from the data. Furthermore, estimates of these parameters can be subject to considerable variability when sample sizes are small. It is thus desirable to incorporate estimation uncertainty into both niche region and overlap calculations. This is naturally achieved within the Bayesian framework discussed by Jackson et al. (2011) and also adopted here.

Let $X_{n\times N} = (X_1, \ldots, X_N)$ now denote a matrix of n-dimensional isotope measurements on N individuals randomly sampled from a given normal distribution $\mathcal{N}_n(\mu, \Sigma)$. By assigning a prior distribution $\pi(\mu, \Sigma)$ to the model parameters, we obtain a posterior distribution $p(\mu, \Sigma \mid X)$. If M samples from $p(\mu, \Sigma \mid X)$ can be obtained, then M ellipses and M values of the overlap metric can be calculated and subsequently used to assess the variability in each of these measures.

As with all Bayesian analyses, several considerations factor into the choice of prior. For sufficiently large samples, the choice of prior is inconsequential. For small sample sizes, including a priori information about the plausible values of μ and Σ can greatly improve the

19399170, 2015, 2, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/14-0235.1 by Universidade Do Algarve, Wiley Online Library on [27/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

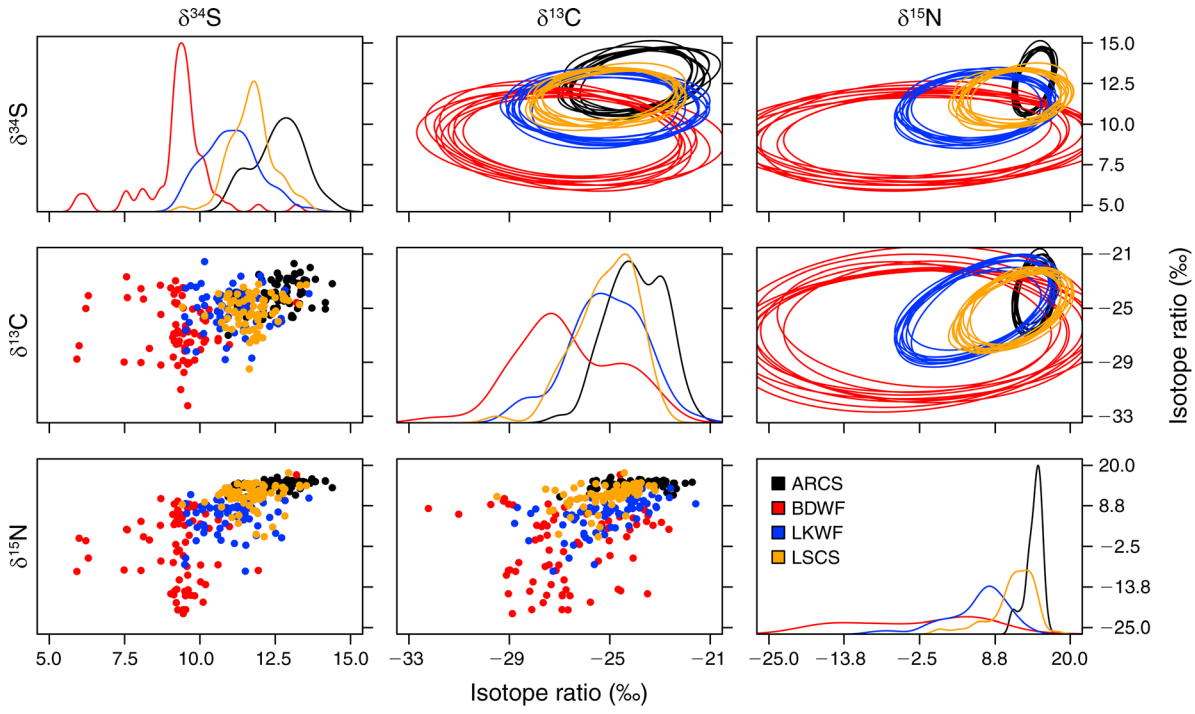


FIG. 3. Ten random elliptical projections of trophic niche region (N_R) for each species and pair of isotopes (elliptical plots). Also displayed are one-dimensional density plots (lines) and two-dimensional scatterplots. The four species displayed are arctic cisco (ARCS), broad whitefish (BDWF), lake whitefish (LKWF), and least cisco (LSCS). Smoothed histograms (density plots) and scatterplots indicate no glaring violation of the assumption of normality.

accuracy of parameter estimates. However, it is also the case with small sample sizes that a poorly specified prior could dominate the analysis and lead to substantial estimation bias. Further discussion on prior selection is included in Appendix C, along with sampling algorithms for $p(\mu, \Sigma \mid X)$ for two different prior considerations. The first applies when prior information is available on the first and second moments of μ and Σ. The second pertains to situations where prior information on μ and Σ is unavailable, vague, or difficult to quantify reliably. In such cases, we suggest a "default" noninformative prior

$$\pi(\mu, \Sigma) \propto |\Sigma|^{-n-1}. \quad (4)$$

For such a prior, the Bayesian point estimates (posterior means) $\hat{\mu}$ and $\hat{\Sigma}$ coincide with the usual frequentist parameter estimates

$$\hat{\mu} = E[\mu \mid X] = \bar{X} = \frac{1}{N}\sum_{n=1}^{N} X_i$$

$$\hat{\Sigma} = E[\Sigma \mid X] = \frac{1}{N-1}\sum_{n=1}^{N}(X_i - \bar{X})(X_i - \bar{X})'. \quad (5)$$

See Appendix C for further discussion of prior specification and for an evaluation of the default prior (Eq. 4) relative to our application.

APPLICATION

We illustrate our niche metrics using data collected from an estuarine area of the Beaufort Sea, at Phillips Bay, Yukon Territory, Canada (69.28° N, 138.49° W). Several fish species, including lake whitefish (*Coregonus clupeaformis*), broad whitefish (*Coregonus nasus*), arctic cisco (*Coregonus autumnalis*), and least cisco (*Coregonus sardinella*) have been collected for monitoring purposes in this area since the 1980s. The data used in the present analysis are from fishes collected in 2007 and 2008. For each of the coregonine species listed, approximately 70 fish were collected and dissected for dorsal muscle tissue. Skinless tissue was analyzed for $\delta^{15}N$, $\delta^{13}C$, and $\delta^{34}S$ ratios at the University of Waterloo Environmental Isotope Laboratory (Waterloo, Ontario, Canada) using established methods (e.g., Power et al. 2012). In this illustrative analysis, we are using raw isotope values: $\delta^{15}N$, $\delta^{13}C$, and $\delta^{34}S$ isotope ratios have not been corrected for baseline, lipid, or fish size (see Appendix D for a data summary).

For each species of fish and every pair of isotopes, 10 random elliptical projections of the trophic niche regions are displayed in Fig. 3. These were obtained by drawing 10 random pairs (μ, Σ) for each species from their posterior distribution, using the default prior (Eq. 4). Fig. 4 displays posterior distributions of the overlap metric $O(^{A}_{B})$. For each pair A and B, random pairs (μ_A, Σ_A) and (μ_B, Σ_B) are drawn from their respective

19399170, 2015, 2, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/14-0235.1 by Universidade Do Algarve, Wiley Online Library on [27/07/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

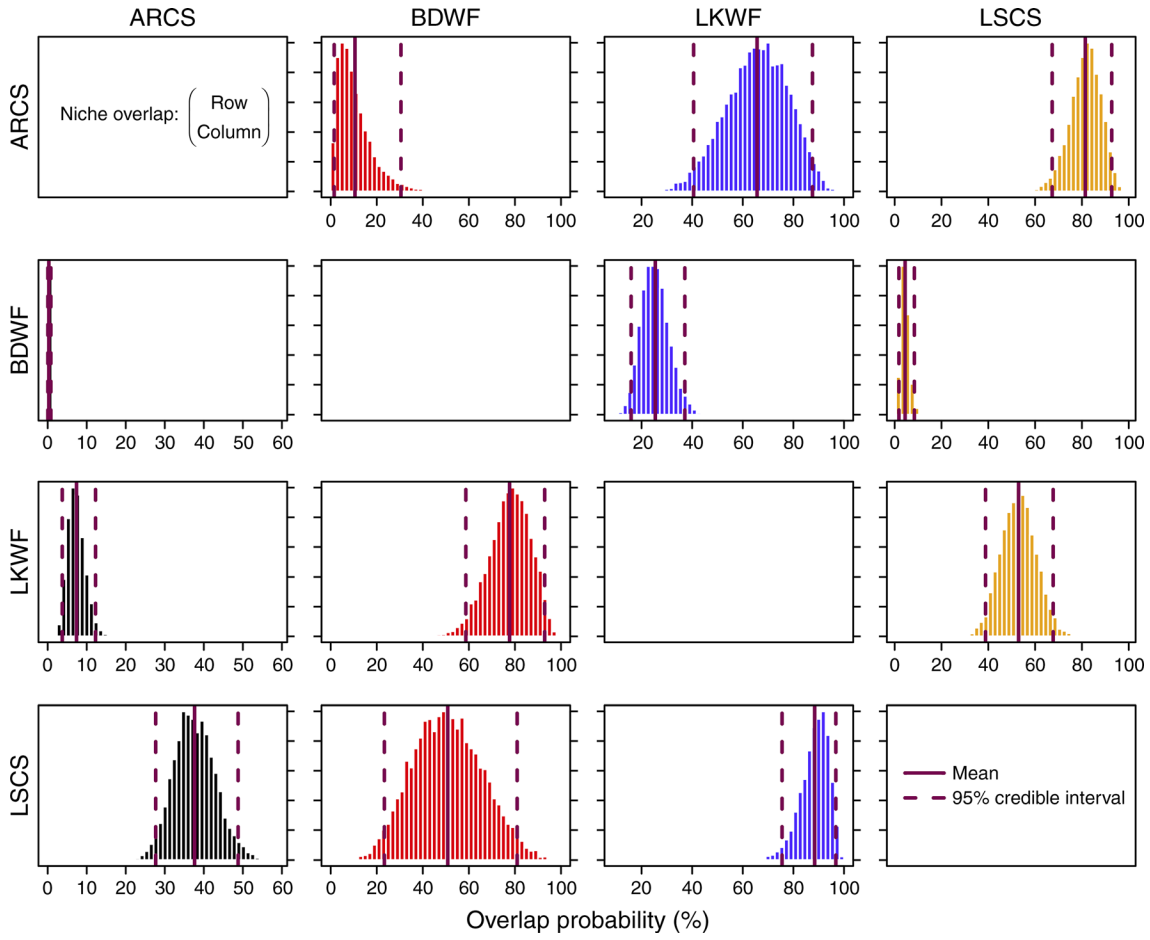


FIG. 4. Posterior distribution of the probabilistic niche overlap metric (%) for a specified N_R of 95% (probability of species displayed in rows overlapping onto those displayed in columns). The posterior means and 95% credible intervals are displayed in purple. Species abbreviations are as in Fig. 3.

distributions. The latter are used to construct a trophic niche region for species *B*, and the former are used to draw 10 000 samples from the normal distribution $\mathcal{N}(\mu_A, \Sigma_A)$ to obtain a single Monte Carlo estimate $\hat{O}(_B^A)$, as in *Statistical methods: Niche overlap*. The entire procedure is repeated 10 000 times to produce the histograms. Posterior means and 95% credible intervals for $O(_B^A)$ are summarized in Appendix D. All code, data, and documentation are available in an R package called nicheROVER (see CRAN web site;[5] R code in Supplement).

RESULTS AND DISCUSSION

The four fish species used in this analysis have anadromous or partially anadromous life histories, and thus were expected to have isotope ratios that indicated use of resources in freshwater, estuarine, and marine environments. Our findings are consistent with this, and consistent with empirical data on habitat use and the life history of coregonids in this region (Reist and Bond 1988). The use of habitats with varying salinity is evident from the large range in $\delta^{34}S$ ratios; broad whitefish showed especially large variability along this axis (Fig. 3). Ratios of $\delta^{13}C$ and $\delta^{15}N$ also varied more widely in broad whitefish than in the other three species, which would obviously contribute to a larger overall N_R (Fig. 3). Consistent with what we know about the ecology of these species (Reist and Bond 1988), arctic cisco appeared to have the smallest N_R, which was due in large part to low variability in $\delta^{34}S$ ratios (Fig. 3).

Conclusions such as these are not the focus of this contribution, but they illustrate an important point. Interpretations of bivariate data (such as in Fig. 3) are almost universal in isotope ecology, and have most commonly involved analyses of $\delta^{15}N$ and $\delta^{13}C$. As we expand the number of isotope tracers employed in ecological studies (e.g., $\delta^{18}O$, δD) and combine isotopic indicators of niche with other indicators of niche (e.g., physical or chemical habitat variables), we will no doubt continue to rely on bivariate projections of multidimen-

[5] http://cran.r-project.org/web/packages/nicheROVER/index.html

Reports

19399170, 2015, 2, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/14-0235.1 by Universidade Do Algarve, Wiley Online Library on [27/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

sional data for interpretation. As shown in Fig. 2, two-dimensional geometric projections of multivariate data suffer from an undesirable dependence on the total number of axes. To use an isotopic example, the definition of the trophic niche region in $\delta^{15}N$–$\delta^{13}C$ space should not depend on whether $\delta^{34}S$ data is included in the analysis, and indeed it does not when using our methodology.

While recent efforts to quantify trophic niche and niche overlap have often been limited to bivariate isotopic data, we are certainly not the first researchers to propose a method for quantifying multidimensional niches and niche overlap (e.g., Green 1971, Pianka 1974, Petraitis 1979). Petraitis (1979) considered a probabilistic approach to niche overlap in a similar spirit as ours. However, the overlap metric therein is based on testing the hypothesis that the resource use in two species is exactly the same. In any application, the value of this overlap metric must converge on zero as more samples are collected.

Also similar to earlier authors, we quantify pairwise niche overlap in a way that accounts for asymmetry in species niche breadths (e.g., Pianka 1974, Petraitis 1979). Our method, however, offers several possible advantages. As well as accommodating $n > 2$ dimensions, it incorporates statistical uncertainty through a Bayesian framework, and does not assume uniform distribution of individuals within the niche region when calculating overlap (see Fig. 1). Furthermore, our method is insensitive to sample size; increasing sample size will not cause spurious increases in niche region (Layman et al. 2007) or decreases in overlap (Petraitis 1979).

Cautions and future directions

As with any method of calculating niche size and overlap, onus is on the researcher to interpret ecological relevance. We found that there was a high probability (>88%) of least cisco overlapping onto the trophic niche of lake whitefish (Fig. 4; Appendix D). To interpret this, we need to consider factors such as resource availability, relative abundance of each species, and which axis or axes is/are contributing most to overlap ($\delta^{13}C$ in this example; Fig. 3). It may be useful to combine the discriminant function analyses employed by Green (1971) with our approach of quantifying niche overlap to better evaluate which axes are driving niche similarities and/or differences.

The level of α used to define N_R also warrants careful consideration by the ecologist (we used 95% and suggest this as a default value). The choice of α will affect the probability of overlap, which will approach either 0% or 100% as α goes to 0 or 1, respectively. In Appendix D, we demonstrate how changing α affects probability of overlap in our example data. If practitioners choose to use an α other than 95%, we recommend presenting a similar analysis to that shown in Appendix D.

Comparing niche size and overlap among habitats or communities is problematic (e.g., Petraitis 1979). Intersite differences in resource availability, community composition, or other factors will affect estimates of niche region and niche overlap. Isotope ecologists are continually faced with the challenge of accounting for spatial and temporal variability in available resources (e.g., Post 2002). If only stable isotope data are being considered, intersite comparisons of N_R and trophic niche overlap may be possible by transforming data into *p*-space (Newsome et al. 2007) or using other normalization procedures. These would require careful consideration by the ecologist.

We quantified N_R with the assumption that our data were multivariate normal. While this is often reasonable for ecological data (sometimes with transformations), there may be cases or variables where the assumption of normality is inappropriate. To meet this particular challenge, it is possible to model the data using a Gaussian copula (e.g., Trivedi and Zimmer 2005). This approach retains the analytic tractability of the multivariate normal while allowing arbitrary specification of each univariate variable distribution.

Concluding remarks

Methods for quantifying species' niches and niche overlap have moved from mostly considering multivariate data ($n > 2$ axes) in the 1960s and 1970s to more recent focus on bivariate data (usually $\delta^{15}N$ and $\delta^{13}C$ isotope ratios). In an effort to realign Hutchinson's (1957) concept of a niche with analytical approaches applied to stable isotope data, we propose a new probabilistic method to quantify niche regions (N_R) and overlap ($O(^{A}_{B})$). Our method can be used on *n*-dimensional continuous data (stable isotope or other), incorporates uncertainty, is directional (asymmetric), does not assume a uniform distribution along a resource gradient, and results in consistent bivariate projections for visual display of data. While there are certainly limitations and improvements to be made, our aim is to build upon the recent advancements of others (e.g., Layman et al. 2007, Jackson et al. 2011), and spark discussion regarding the alignment of analytical approaches with the theoretical niche concepts they are used to investigate.

Acknowledgments

Funding for sample collection and analysis was provided by the Natural Sciences and Engineering Research Council (to M. Power), ArcticNet (M. Power), the Arctic Coastal Ecosystem Studies program, Northern Coastal Marine Studies Program, and the Nahidik program. H. K. Swanson and M. Lysy contributed equally to this work.

Literature Cited

Araújo, M. S., D. I. Bolnick, G. Machado, A. A. Giaretta, and S. F. dos Reis. 2007. Using $\delta^{13}C$ stable isotopes to quantify individual-level diet variation. Oecologia 152(4):643–654.

Bearhop, S., C. E. Adams, S. Waldron, R. A. Fuller, and H. MacLeod. 2004. Determining trophic niche width: a novel approach using stable isotope analysis. Journal of Animal Ecology 73(5):1007–1012.

19399170, 2015, 2, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/14-0235.1 by Universidade Do Algarve, Wiley Online Library on [27/07/2023]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Bolnick, D. I., R. Svanbäck, J. A. Fordyce, L. H. Yang, J. M. Davis, C. D. Hulsey, and M. L. Forister. 2003. The ecology of individuals: incidence and implications of individual specialization. American Naturalist 161(1):1–28.

Box, G. E., and G. Tiao. 1973. Bayesian inference in statistical analysis. Addison-Wesley, Reading, Massachusetts, USA.

Colwell, R. K., and D. J. Futuyma. 1971. On the measurement of niche breadth and overlap. Ecology 52:567–576.

Gause, G. F., O. K. Nastukova, and W. W. Alpatov. 1934. The influence of biologically conditioned media on the growth of a mixed population of *Paramecium caudatum* and *P. aureliax*. Journal of Animal Ecology 3(2):222–230.

Green, R. H. 1971. A multivariate statistical approach to the Hutchinsonian niche: bivalve molluscs of central Canada. Ecology 52:544–556.

Guzzo, M. M., G. D. Haffner, N. D. Legler, S. A. Rush, and A. T. Fisk. 2013. Fifty years later: trophic ecology and niche overlap of a native and non-indigenous fish species in the western basin of Lake Erie. Biological Invasions 15:1695–1711.

Hutchinson, G. E. 1957. Concluding remarks. Cold Spring Harbor Symposia on Quantitative Biology 22:415–427.

Jackson, A. L., R. Inger, A. C. Parnell, and S. Bearhop. 2011. Comparing isotopic niche widths among and within communities: SIBER stable isotope Bayesian ellipses in R. Journal of Animal Ecology 80:595–602.

Jackson, M. C., I. Donohue, A. L. Jackson, J. R. Britton, D. M. Harper, and J. Grey. 2012. Population-level metrics of trophic structure based on stable isotopes and their application to invasion ecology. PLoS ONE 7(2):e31757.

Layman, C. A., D. A. Arrington, C. G. Montaña, and D. M. Post. 2007. Can stable isotope ratios provide for community-wide measures of trophic structure? Ecology 88:42–48.

Moore, J. W., and B. X. Semmens. 2008. Incorporating uncertainty and prior information into stable isotope mixing models. Ecology Letters 11(5):470–480.

Newsome, S. D., C. Martinez del Rio, S. Bearhop, and D. L. Phillips. 2007. A niche for isotopic ecology. Frontiers in Ecology and the Environment 5:429–436.

Parnell, A. C., and A. L. Jackson. 2013. Stable isotope analysis in R (SIAR), v. 4.2. http://cran.r-project.org/web/packages/siar/

Petraitis, P. S. 1979. Likelihood measures of niche breadth and overlap. Ecology 60:703–710.

Phillips, D. L., and J. W. Gregg. 2003. Source partitioning using stable isotopes: coping with too many sources. Oecologia 136(2):261–269.

Pianka, E. R. 1974. Niche overlap and diffuse competition. Proceedings of the National Academy of Sciences USA 71(5):2141–2145.

Post, D. M. 2002. Using stable isotopes to estimate trophic position: models, methods, and assumptions. Ecology 83:703–718.

Power, M., M. O'Connell, and J. Dempson. 2012. Determining the consistency of thermal habitat segregation within and among arctic charr morphotypes in Gander Lake, Newfoundland. Ecology of Freshwater Fish 21:245–254.

Reist, J., and W. Bond. 1988. Life history characteristics of migratory coregonids of the lower Mackenzie River, Northwest Territories, Canada. Finnish Fisheries Research 9:133–144.

Schmidt, S. N., M. J. Vander Zanden, and J. F. Kitchell. 2009. Long-term food web change in Lake Superior. Canadian Journal of Fisheries and Aquatic Sciences 66(12):2118–2129.

Trivedi, P. K., and D. M. Zimmer. 2005. Copula modeling: an introduction for practitioners. Foundations and Trends in Econometrics 1(1):1–111.

Supplemental Material

Ecological Archives

Appendices A–D and the Supplement are available online: http://dx.doi.org/10.1890/14-0235.1.sm

19399170, 2015, 2, Downloaded from https://esajournals.onlinelibrary.wiley.com/doi/10.1890/14-0235.1 by Universidade Do Algarve, Wiley Online Library on [27/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License